

Learning on Adaptive Manifolds for Graph Collaborative Filtering

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Abstract

Graph-based collaborative filtering has advanced by modeling higher-order interactions, yet performance remains constrained by underlying geometric assumptions and propagation schemes. User-item interaction graphs typically exhibit pronounced topological heterogeneity, whereas existing methods rely on a fixed, homogeneous geometry and employ tangent space aggregation. To address these fundamental limitations, this paper introduces Adaptive Geometric Collaborative Filtering (AGCF), a novel method rooted in Hamiltonian dynamics, which reframes representation learning as a physical process evolving on a time-varying manifold. AGCF is distinguished by an integrated design comprising: (1) a learnable, node-dependent Riemannian metric that construct a continuous heterogeneous manifold aligned with local topology; (2) unified dynamic trajectories that achieve intrinsic propagation without tangent space approximations; (3) a channel-wise metric that captures semantic anisotropy in the feature space. We rigorously prove global existence and uniqueness of the induced dynamics and explain the mechanism enabling long-range information propagation. Extensive experiments on five benchmark datasets show consistent gains over representative baselines.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Recommender System; Graph Collaborative Filtering; Riemannian Manifold

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1 Introduction

In the vast landscape of information, recommender systems [7, 9, 11, 29, 35, 40, 51, 55] have become critical infrastructure connecting users with content, products, and services. The core task

Table 1: Comparison of AGCF with relevant methods. Our model is the first to simultaneously address all three fundamental limitations.

Method	Heterogeneous Manifold	Intrinsic Propagation	Semantic Anisotropy
LightGCN [22]	X	✓	X
HGCF [41]	X	X	X
HICF [49]	X	X	X
GDCF [56]	X	-	X
HamGNN [25]	✓	X	✓
AGCF	✓	✓	✓

is to learn generalizable preference patterns from historical user-item interactions, enabling efficient retrieval and ranking. As the most representative technical approach, Collaborative Filtering (CF) [23, 24, 27, 28, 30, 34] has recently been advanced by Graph Neural Networks (GNNs) [12, 21, 26, 33, 42, 43, 46, 48, 52], which explicitly model user-item relationships as bipartite graphs. By aggregating information from multi-hop neighbors, GNNs have significantly improved recommendation performance [13, 22, 36, 37, 44, 45]. Yet, despite increasing sophistication, the performance ceiling remains limited by assumptions about geometry and by the chosen propagation schemes.

This paper aims to challenge and overcome three of the most critical limitations. **The first and most fundamental is geometric homogeneity.** Existing methods, from classic Euclidean models to more advanced non-Euclidean approaches, typically learn representations within a geometrically homogeneous space. The flat geometry of Euclidean space struggles to embed the hierarchical structures prevalent in user-item interaction graphs with low distortion [3, 5, 31, 38]. To address this, hyperbolic spaces are introduced and have demonstrated superior performance in capturing such hierarchies [41, 47, 49, 50, 54]. However, enforcing a single global curvature remains inadequate for graphs with complex topological heterogeneity. A typical user-item network is rarely a pure tree; it often intertwines multiple topological patterns, such as tree-like structures radiating from popular entities, dense communities formed by users with highly consistent interests, and cycles of substitutability composed of functionally similar items. Subsequent solutions like product manifolds [56], while acknowledging the existence of multiple topologies, still apply a fixed, globally shared set of geometric rules to all nodes. This overlooks a key fact: the geometry of a user-item interaction graph is local and spatially varying, rather than globally uniform. Thus, an effective model should be capable of constructing an adaptive heterogeneous manifold.

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The second critical limitation stems from the reliance on tangent space for aggregation. To perform neighbor aggregation on Riemannian manifolds, common practice follows a three-step paradigm: neighbor representations are projected from the manifold to a tangent space at a reference point via the logarithmic map; aggregation is performed within this temporary, flat Euclidean space; the result is mapped back to the manifold using the exponential map. This routine introduces a conceptual decoupling and approximation distortion between the geometry and the propagation process. Each mapping may incur information loss, and the aggregation is confined to a universal tangent space that ignores the local context of the node. A more unified design should realize intrinsic propagation directly on the manifold so that every step of information flow is guided by local geometry.

The third, widely overlooked limitation is semantic isotropy. In the feature space, many models implicitly assume all semantic directions are equivalent. This appears in the use of inner products or Euclidean distances, which weigh dimensions equally and overlook inter-dimensional correlations. A stronger formulation should not only understand which users and items are close on the graph but also along which meaningful semantic directions they should be oriented to achieve a more precise and discriminative alignment.

The landscape of representative methods, summarized in Table 1, reveals a systemic bottleneck. Although advanced concepts like heterogeneous manifolds have appeared in general graph learning (e.g., HamGNN [25]), to the best of our knowledge, no collaborative filtering model simultaneously offers a continuously adaptive geometry, a fully intrinsic propagation mechanism, and the capacity to model semantic anisotropy. This highlights a clear and significant gap in the literature.

To break through this impasse, we propose Adaptive Geometric Collaborative Filtering (AGCF), a novel approach that couples adaptive geometric modeling with semantic-aware intrinsic propagation under a unified dynamical perspective. Instead of viewing nodes as static points, we model them as particles moving on a dynamically evolving manifold, governed by Hamiltonian mechanics. AGCF systematically addresses the three challenges with an integrated design. First, a node-level learnable matrix elevates local geometry from a global constant to a position-dependent Riemannian metric, breaking geometric homogeneity. Second, unified dynamic trajectories enable information to propagate entirely within the manifold, avoiding decoupling from tangent space projections; theoretical analysis further shows that this dynamic process let higher-order neighborhoods naturally influence the central node over successive updates, enabling long-range collaborative effects. Third, a channel-wise metric defines anisotropy in the feature space, learning correlations and relative importance across dimensions to guide more meaningful interactions.

A theoretical foundation supports the system: global existence, uniqueness, and energy boundedness of the solutions are established, ensuring stability during training.

The main contributions are as follows:

- **Heterogeneous Manifold Framework for Collaborative Filtering.** Through a node-dependent learnable Riemannian

metric, AGCF constructs a heterogeneous manifold that adaptively aligns with local interaction topologies, overcoming geometric homogeneity in prior methods.

- **Unified Intrinsic Propagation Mechanism.** Dynamic trajectories realize direct propagation on the manifold, avoiding tangent space approximations and capturing long-range collaborative signals continuously.
- **Semantic Anisotropy.** A channel-wise metric models correlations and importance among feature dimensions, yielding more precise and semantically discriminative representation alignment.
- **Theoretical Guarantees and Empirical Validation.** Global existence and uniqueness of the proposed dynamics system are proved, and the long-range propagation mechanism is explained. Extensive experiments on multiple benchmark datasets validate the consistent improvements.

2 Related Work

2.1 Graph Collaborative Filtering

Graph Neural Networks (GNNs) have become the dominant paradigm in modern Collaborative Filtering (CF). NGCF [44] first introduces graph convolutions to CF, while LightGCN [22] is an exceptionally strong Euclidean baseline by simplifying the GCN architecture. Motivated by the hierarchical structure of user-item interaction graphs, recent works attempt to extend CF models to hyperbolic space, with HGCF [41] being a pioneering effort. Building upon it, subsequent research has introduced various improvements: HICF [49] proposes an adaptive margin loss to better balance head and tail items; HGCH [54] employs an intrinsic gyromidpoint for aggregation and explored the fusion of side information. Furthermore, GDGC [47] captures diverse user intents in a mixed-curvature space.

2.2 Riemannian Graph Neural Networks

Hyperbolic GNNs [4–6, 8, 10, 14, 32, 53, 57, 59] have achieved great success in modeling hierarchical structures due to the exponential growth of their volume with radius. To handle more complex topological patterns, subsequent works begin to explore mixed-curvature spaces, typically by constructing a global product manifold from the Cartesian product of multiple constant-curvature spaces (Euclidean, hyperbolic, spherical) [1, 17]. Nonetheless, the aforementioned methods mostly rely on a homogeneous geometric background, where all nodes are subject to the same globally fixed geometric rules. To address the topological heterogeneity of graphs, more advanced works have started to explore adaptive geometries. For instance, GraphMoRE [18] employs a Mixture-of-Experts architecture to select from a predefined, discrete set of expert geometries for each node. HamGNN [25] makes the attempt at constructing a continuously adaptive geometry for general graph tasks through a dynamical framework.

2.3 Hamiltonian Neural Networks

Hamiltonian Neural Networks (HamNNs) are an emerging research area that aims to incorporate principles from Hamiltonian mechanics as an inductive bias into deep learning. Foundational work [19]

provided key theoretical support for the training stability of such dynamical architectures, demonstrating that gradient issues can be mitigated through structural design. In the practical applications, Greydanus et al. [16] took the lead in applying HamNNs to the field of physics. Subsequent work further integrated Hamiltonian mechanics with graph networks [2].

3 Preliminaries

3.1 Riemannian Manifolds

In geometry, a manifold $\mathcal{M}$ is a topological space that locally resembles a Euclidean space. A d -dimensional manifold can thus be seen as a space that is locally flat but may be globally curved. At each point $q \in \mathcal{M}$, there is an associated d -dimensional vector space, known as the tangent space $\mathcal{T}_q\mathcal{M}$. A vector $v \in \mathcal{T}_q\mathcal{M}$ can be intuitively understood as the velocity of a curve passing through q . Dually, there exists a cotangent space $\mathcal{T}_q^*\mathcal{M}$, whose elements $p \in \mathcal{T}_q^*\mathcal{M}$ are called covectors. In classical mechanics, covectors typically correspond to the generalized momentum of a system.

A Riemannian manifold, denoted as $(\mathcal{M}, g)$, is a smooth manifold $\mathcal{M}$ equipped with a Riemannian metric g . The metric g defines an inner product $\langle \cdot, \cdot \rangle_{g(q)} : \mathcal{T}_q\mathcal{M} \times \mathcal{T}_q\mathcal{M} \rightarrow \mathbb{R}$ on each tangent space, which varies smoothly with the point q . The inner product of any two tangent vectors $u, v \in \mathcal{T}_q\mathcal{M}$ is given in local coordinates by the $d \times d$ symmetric positive-definite matrix $g(q)$:

$$\langle u, v \rangle_{g(q)} = u^\top g(q) v. \quad (1)$$

The length of a tangent vector u is then defined as $\|u\|_{g(q)} := \sqrt{\langle u, u \rangle_{g(q)}}$. The metric matrix $g(q)$ determines local geometric quantities such as distances, angles, and curvature.

3.2 Geodesics and the Exponential Map

On a Riemannian manifold, the natural generalization of the straight line is the geodesic. A geodesic is the locally shortest path between points and can be found by minimizing the energy functional $E(\gamma)$ of a curve:

$$E(\gamma) = \frac{1}{2} \int_a^b \|\dot{\gamma}(t)\|_{g(\gamma(t))}^2 dt \quad (2)$$

where $\gamma : [a, b] \rightarrow \mathcal{M}$ is a curve and $\dot{\gamma}(t)$ is its velocity vector.

Based on geodesics, one can define the exponential map, $\exp_q(\cdot) : \mathcal{T}_q\mathcal{M} \rightarrow \mathcal{M}$, which maps a vector v in the tangent space to the point on the unique geodesic starting at q with initial velocity v , after a unit of time. Formally, we have

$$\exp_q(v) = \gamma(1), \quad (3)$$

where $\gamma(0) = q$ and $\dot{\gamma}(0) = v$.

3.3 Hamiltonian Dynamics

Hamiltonian System. Hamiltonian mechanics provides a general framework for describing the evolution of dynamical systems. The state of a system is described by its generalized coordinates q (a position vector) and generalized momenta p (a covector). This pair (q, p) constitutes a point in the phase space. Mathematically, this $2d$ -dimensional phase space is the cotangent bundle $T^*\mathcal{M}$ of the manifold $\mathcal{M}$.

The evolution of the system is governed by a scalar function called the Hamiltonian $H(q, p)$, which often represents the total energy. The system evolves according to Hamilton's Equation:

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q} \quad (4)$$

The trajectory traced by the solution $(q(t), p(t))$ is called a Hamiltonian orbit.

Geodesic Flow. The geodesic flow, the dynamical process in which particles traverse geodesics, can be precisely described as a specific Hamiltonian system.

Specifically, when the Hamiltonian of a system consists solely of the kinetic energy term defined by the Riemannian metric g (i.e., potential energy $U(q) = 0$):

$$H(q, p) = K(q, p) = \frac{1}{2} p^\top g(q)^{-1} p \quad (5)$$

where $g(q)^{-1}$ is the inverse matrix of the metric matrix $g(q)$, the projection of the resulting Hamiltonian orbit $(q(t), p(t))$ onto $\mathcal{M}$, the curve $q(t)$, is precisely a geodesic on the manifold.

This crucial connection reveals that Hamiltonian orbits are a strict generalization of geodesics. Geodesics describe free motion in the absence of a potential field, whereas a more general Hamiltonian with a non-zero potential energy term $U(q)$ can describe far richer evolutionary trajectories within complex force fields, such as those induced by the structure of a graph. This provides a unified theoretical language for the subsequent modeling of complex node interactions.

4 Methodology

This section details the proposed Adaptive Geometric Collaborative Filtering (AGCF) method. We consider a standard implicit feedback setting for collaborative filtering with m users, $\mathcal{U} = \{u_1, \dots, u_m\}$, and n items, $\mathcal{I} = \{i_1, \dots, i_n\}$. The historical interactions are recorded in a sparse $m \times n$ binary matrix $\mathbf{R}$, where $R_{ui} = 1$ if user u has interacted with item i , and $R_{ui} = 0$ otherwise. These interactions naturally form a bipartite graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where the set of vertices is $\mathcal{V} = \mathcal{U} \cup \mathcal{I}$ and the edge set is $\mathcal{E} = \{(u, i) | R_{ui} = 1\}$. Our goal is to learn a d -dimensional embedding vector $q_v \in \mathbb{R}^d$ for each node $v \in \mathcal{V}$ (i.e., for all users and items), and to use these embeddings to predict user preferences for unobserved items.

The core idea is to model the embedding learning process for all nodes as a system evolving in continuous time, governed by Hamiltonian dynamics. We regard the state of each node as a pair of vectors in a phase space: a position q_v and a momentum p_v . The evolution of this pair is governed by a dynamical equation composed of a conservative Hamiltonian energy function and a dissipative damping term.

4.1 Hamiltonian Formulation

The dynamical system is defined by a Hamiltonian H , which represents the total energy of the system and is composed of a kinetic term K and a potential term U .

State Initialization. We first map all users and items in the graph $\mathcal{G}$ into a d -dimensional latent space via a standard embedding layer, yielding a learnable position embedding matrix $\mathbf{Q} \in \mathbb{R}^{N \times d}$, where $N = m + n$ is the total number of nodes. To describe the full

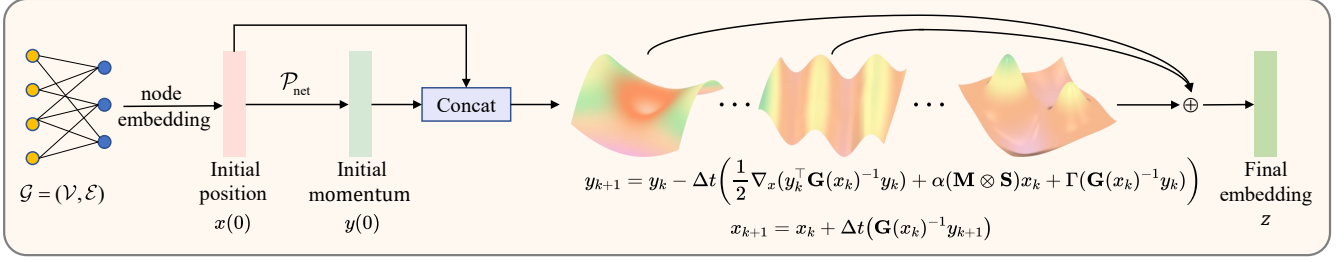


Figure 1: The overall architecture of AGCF. The node states (x, y) evolve along a continuous trajectory governed by Hamiltonian equations. Crucially, the underlying geometric manifold (defined by the metric $G(x)$) is dynamic and adaptive, continuously changing its shape as the node positions $x(t)$ evolve.

dynamics of the system, we introduce a corresponding momentum vector for each node, which form the momentum matrix $\mathbf{P} \in \mathbb{R}^{N \times d}$. The initial momentum $\mathbf{P}(0)$ is not an independent parameter but is generated from the initial positions $\mathbf{Q}(0)$ via a neural network, $\mathcal{P}_{\text{net}}$:

$$\mathbf{P}(0) = \mathcal{P}_{\text{net}}(\mathbf{Q}(0)). \quad (6)$$

For notational convenience in the following derivations, we denote the flattened vector forms of the position and momentum matrices as $x = \text{vec}(\mathbf{Q})$ and $y = \text{vec}(\mathbf{P})$, respectively, where $x, y \in \mathbb{R}^{Nd}$.

Kinetic Energy and Adaptive Geometry. The position dependent Riemannian metric of the system is implemented as a large $Nd \times Nd$ block-diagonal matrix $\mathbf{G}(x)$. This matrix, $\mathbf{G}(x)$, serves as the concrete implementation for the entire coupled system of the abstract metric $g(q)$ in the equation 5. It takes the form $\mathbf{G}(x) = \text{blockdiag}(G_1(x_1), \dots, G_N(x_N))$, where each block $G_v(x_v)$ is a $d \times d$ symmetric positive-definite matrix produced by a neural network that takes the position vector x_v of node v as input. Based on this metric, the kinetic energy of the system is defined as:

$$K(x, y) = \frac{1}{2} y^\top \mathbf{G}(x)^{-1} y. \quad (7)$$

Potential Energy and Anisotropic Interactions. The potential energy term defines the interaction forces between nodes guided by the graph structure. It is defined as a quadratic form over the position vectors x :

$$U(x) = \frac{1}{2} x^\top (\mathbf{M} \otimes \mathbf{S}) x. \quad (8)$$

The core of this potential is the system coupling matrix $\mathbf{M} \otimes \mathbf{S}$, formed by the Kronecker product of two components. The first is the $N \times N$ **structural matrix** $\mathbf{M} = (1 + \delta)\mathbf{I} + \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$, where $\delta > 0$ is small, $\mathbf{D}$ is the degree matrix and $\mathbf{A}$ is the adjacency matrix of the bipartite graph, defined as:

$$\mathbf{A} = \begin{pmatrix} \mathbf{0}_{m \times m} & \mathbf{R} \\ \mathbf{R}^\top & \mathbf{0}_{n \times n} \end{pmatrix}. \quad (9)$$

The second component is the $d \times d$ **channel-wise metric** $\mathbf{S}$, a symmetric positive-definite matrix defining the semantic anisotropy. The Kronecker product $\mathbf{M} \otimes \mathbf{S}$ results in an $Nd \times Nd$ block matrix

of the form:

$$\mathbf{M} \otimes \mathbf{S} = \begin{pmatrix} \mathbf{M}_{11} \mathbf{S} & \dots & \mathbf{M}_{1N} \mathbf{S} \\ \vdots & \ddots & \vdots \\ \mathbf{M}_{N1} \mathbf{S} & \dots & \mathbf{M}_{NN} \mathbf{S} \end{pmatrix}. \quad (10)$$

Equations of Motion. The total energy of the system is defined by the Hamiltonian $H(x, y) = K(x, y) + \alpha U(x)$, where $\alpha > 0$ is a hyperparameter balancing the two terms. The evolution of the node embeddings is then governed by the damped Hamilton's equations:

$$\dot{x} = \frac{\partial H}{\partial y}, \quad \dot{y} = -\frac{\partial H}{\partial x} - \Gamma \frac{\partial H}{\partial y}, \quad (11)$$

where Γ is a positive semi-definite damping matrix that ensures the system converges to a stable final state.

The specific form of these equations is determined by the gradients of the Hamiltonian H . First, the velocity vectors of all nodes in the system are given by:

$$\dot{x} = \frac{\partial H}{\partial y} = \mathbf{G}(x)^{-1} y. \quad (12)$$

The rate of change of the momentum y is driven by both conservative forces (from $-\partial H / \partial x$) and dissipative forces (from the damping term). The conservative force itself can be further decomposed into:

$$-\frac{\partial H}{\partial x} = -\frac{1}{2} \nabla_x (y^\top \mathbf{G}(x)^{-1} y) - \alpha (\mathbf{M} \otimes \mathbf{S}) x. \quad (13)$$

The first term is the geometric force, arising from the variation of the geometry of the manifold, while the second is the interaction force, determined by the graph topology and semantic anisotropy.

Combining all terms, the final dynamical equations for the AGCF system are obtained:

$$\begin{aligned} \dot{x} &= \mathbf{G}(x)^{-1} y, \\ \dot{y} &= -\frac{1}{2} \nabla_x (y^\top \mathbf{G}(x)^{-1} y) - \alpha (\mathbf{M} \otimes \mathbf{S}) x - \Gamma \mathbf{G}(x)^{-1} y. \end{aligned} \quad (14)$$

This set of equations elegantly unifies all our design principles, clearly demonstrating how the evolution of node embeddings is co-regulated by local geometry, graph interactions, and system damping. In practice, the complex geometric force term $\nabla_x(\cdot)$ can be computed accurately by automatic differentiation frameworks without manual derivation.

4.2 Implementation

Parameterizing the Metric Matrices. As the kinetic energy term only depends on the inverse of the Riemannian metric $G(x)$, the implementation opts to parameterize the inverse metric directly to enhance computational efficiency. All metric matrices in the framework must be symmetric positive-definite (SPD). To enforce this property, for each node v , a MLP takes its position vector x_v as input to generate an unconstrained $d \times h$ matrix A_v ($h \ll d$), and constructs:

$$G_v(x_v)^{-1} = A_v A_v^\top + \epsilon I, \quad (15)$$

where ϵ is a small positive constant. The global channel-wise metric S is parameterized analogously.

Numerical Integration. For the damping matrix Γ , an isotropic damping form is adopted, setting $\Gamma = \gamma I$, where $\gamma > 0$ is a scalar hyperparameter. Our dynamical system is composed of a dominant, conservative Hamiltonian part and a dissipative damping part. To respect this mixed conservative-dissipative structure upon discretization and achieve long-term stability, Geometric Numerical Integration is more robust than conventional methods [15]. To this end, this work employs the first-order Symplectic Euler method [20], which preserves the symplectic structure of the conservative Hamiltonian component. Higher-order methods, such as the Störmer-Verlet integrator, would significantly increase computational complexity.

The evolution from $t = 0$ to a terminal time T is discretized into K steps, with a step size of $\Delta t = T/K$. The one-step update rule for the Symplectic Euler method is as follows:

$$y_{k+1} = y_k - \Delta t \left(\frac{1}{2} \nabla_x (y_k^\top G(x_k)^{-1} y_k) + \alpha (M \otimes S) x_k + \Gamma (G(x_k)^{-1} y_k) \right), \quad (16)$$

$$x_{k+1} = x_k + \Delta t \left(G(x_k)^{-1} y_{k+1} \right), \quad (17)$$

where x_k and y_k denote the position and momentum at discrete time step k .

Final Embedding Generation. Inspired by the layer aggregation strategies in modern graph collaborative filtering models, we form the final node representation by aggregating states from multiple points in time along the dynamic trajectory.

Specifically, we define a set of $L + 1$ output time steps, $\mathcal{T} = \{T_0, T_1, \dots, T_L\}$, where each $T_0 = 0$ and $T_L = LT$, i.e., $T_l - T_{l-1} = T$. The dynamical system evolves from $t = 0$ via numerical integration. At each specified output time $T_l \in \mathcal{T}$, we record the position state of the system, $x(T_l)$. The final node representation z is formed as:

$$z = \sum_{l=0}^L x(T_l), \quad (18)$$

which fuses multi-scale information from the initial identity to deeply evolved states and is then used for the downstream recommendation task.

4.3 Theoretical Analysis

This section provides theoretical guarantees for the AGCF framework, focusing on the global existence and uniqueness of its solutions, as well as its intrinsic mechanism for long-range information

propagation. We first state the central theorems of our system under some mild assumptions:

Assumptions (H1).

- (H1.a) There exist constants $0 < m \leq M < \infty$ such that for any position x in any compact set, the metric matrix $G(x)$ satisfies $mI \leq G(x) \leq MI$.
- (H1.b) The metric matrix $G(x)$ is C^1 -smooth, and its first derivative $\partial G(x)/\partial x$ is bounded and Lipschitz continuous on any compact set.

THEOREM 1 (GLOBAL EXISTENCE AND UNIQUENESS). *Under Assumptions (H1), for any initial condition, the unique solution to the dynamical system AGCF (14) exists for all time $t \in [0, \infty)$.*

This theorem guarantees a well defined forward dynamics, which supports stable training. A detailed proof is provided in Appendix A.

To gain analytical insight into the information propagation mechanism of this framework, we analyze a special case where the metric matrix G is constant. Under this simplification, the equation of motion can be exactly rewritten as a second-order linear ordinary differential equation:

$$\ddot{x}(t) + D\dot{x}(t) + \Omega x(t) = 0, \quad (19)$$

where $D = G^{-1}\Gamma$ and $\Omega = \alpha G^{-1}(M \otimes S)$.

THEOREM 2 (LONG-RANGE INFORMATION PROPAGATION). *For the second-order linear system above, as the dynamical evolution time t increases, the representation of a node becomes progressively and increasingly influenced by the initial states of higher-order neighbors in the graph.*

This theorem, though derived in a simplified linear system, provides profound physical intuition for the behavior of our full model. It provides deep insight into how the dynamical system intrinsically and automatically achieves long-range information propagation. Unlike traditional GNNs that enlarge their receptive fields by discretely stacking graph convolution layers, AGCF naturally fuses information from local to global scales through a single, continuous-time evolutionary process.

4.4 Optimization

We employ the margin ranking loss to optimize the model. Formally, for each training triplet (u, i, j) , where i is a positive item and j is a negative item, the loss function is defined as:

$$\mathcal{L}(u, i, j) = \max(0, d(z_u, z_i)^2 - d(z_u, z_j)^2 + m), \quad (20)$$

where $m > 0$ is a margin hyperparameter.

The key of the loss function above lies in the definition of the distance metric $d(\cdot, \cdot)$. To reflect the fundamental design principle of AGCF regarding semantic anisotropy in the optimization objective, the standard Euclidean distance is not used. Instead, we employ the distance induced by the channel-wise metric S :

$$d(z_u, z_i)^2 := d_S(z_u, z_i)^2 = (z_u - z_i)^\top S (z_u - z_i). \quad (21)$$

This distance metric ensures that the optimization process can directly leverage the information about feature dimension correlations learned by S .

Table 2: Statistics of the five experimental datasets.

Dataset	User	Item	Interactions	Density
MovieLens	6,039	3,308	835,789	4.183%
Amazon-CD	113,303	82,910	1,397,717	0.015%
Amazon-Book	118,039	98,793	2,836,041	0.024%
Gowalla	64,115	164,532	2,018,421	0.019%
Yelp	202,012	86,880	3,022,685	0.017%

During inference, the preference score of a user u for a candidate item i is given by the negative of their distance, ensuring that closer items receive higher preference scores:

$$\text{score}(u, i) = -d_S(x_u, x_i)^2. \quad (22)$$

The time complexity of our model is $O(Nd^2h + |\mathcal{E}|d)$.

5 Experiments

In this section, we conduct extensive experiments to evaluate the effectiveness of AGCF. The experiments are designed to answer the following research questions (RQs):

- RQ1:** How does AGCF perform compared to a range of representative baseline models?
- RQ2:** Does each proposed component of AGCF make a necessary contribution to its final performance?
- RQ3:** How do different formulations of kinetic energy function K affect the performance of AGCF?
- RQ4:** How does AGCF capture long-range collaborative signals as the dynamics evolve?
- RQ5:** How does the damping coefficient γ influence the effectiveness of AGCF?

5.1 Experimental Setup

Datasets and Evaluation Metrics. Our experimental evaluation is conducted on five widely-used public benchmark datasets that vary in domain, scale, and sparsity: MovieLens¹, Amazon-CD², Amazon-Book, Gowalla³, and Yelp⁴ (statistics in Table 2). Experiments are implemented via RecBole [58] using a random 8:1:1 per-user split. We adopt the all-ranking protocol and report Recall@ K and NDCG@ K for $K \in \{10, 20\}$.

Baselines. To ensure a comprehensive evaluation, we compare AGCF against a suite of representative baselines. These include the classic latent factor model: BPRMF [39]; mainstream Euclidean graph collaborative filtering models: NGCF [44] and LightGCN [22]; and a variety of advanced non-Euclidean methods, including hyperbolic models: HGCF [41], HRCF [50], HICF [49], and HGCH [54]; a mixed-curvature model: GDCF [56]; and general graph neural networks designed for topological heterogeneity: HamGNN [25] and GraphMoRE [18].

Hyperparameter Settings. All methods are optimized with Adam using an embedding dimension of 64. We train for a maximum of 500 epochs, employing early stopping based on Recall@10 with a

patience of 30 epochs. Hyperparameters, including the learning rate, L2 regularization weight decay, margin m , and dynamics-related parameters (output time steps L , integration steps K , potential energy strength α , and damping coefficient γ), are tuned via grid search on the validation set, while the evolution time T is fixed at 1.0.

5.2 Overall Performance Comparison (RQ1)

Table 3 reports the detailed performance comparison of AGCF against a suite of representative baselines across five benchmark datasets. The main observations are as follows:

- AGCF consistently attains the best performance across all datasets and metrics, and this uniform lead spans domains, scales, and sparsity regimes, which together indicate strong effectiveness and broad generality.
- Adaptive heterogeneous geometry confers clear advantages. After adaptation to the recommendation setting, GraphMoRE and HamGNN outperform methods that assume homogeneous geometry, including Euclidean, single curvature hyperbolic, and mixed curvature models, across all tested scenarios, suggesting that modeling topological heterogeneity improves representation quality.
- Within the demonstrably superior paradigm of adaptive geometries, AGCF further emerges as the clear leader, consistently surpassing both GraphMoRE and HamGNN. This suggests that the unified dynamical design that combines a node-dependent Riemannian metric, intrinsic propagation, and a channel wise metric in one coherent formulation yields a more fundamental advantage for collaborative filtering than approaches that address only a subset of these challenges.

5.3 Ablation Study (RQ2)

To answer RQ2 and quantify the contribution of each core component, three variants of AGCF are evaluated, each removing one element to isolate its effect:

- **AGCF w/o G:** replaces the node-dependent metric $G(x)$ with a learnable, but globally shared, metric matrix G_{global} .
- **AGCF w/o M:** removes the intrinsic propagation mechanism, instead performing a single GNN-style neighbor aggregation after the terminal time T .
- **AGCF w/o S:** fixes the channel-wise metric S to be the identity matrix I .

Table 4 shows that removing any of the three components leads to a consistent decrease in performance across all datasets and metrics, which indicates that each element is beneficial and necessary for strong results. Among the three, eliminating the adaptive metric produces the largest decline, and this observation supports the view that geometric homogeneity is a primary bottleneck while a node-level heterogeneous manifold is critical to overcoming it. Furthermore, removing intrinsic propagation and the channel wise metric also yields substantial losses, which validates their roles in fusing collaborative signals and achieving fine grained semantic alignment, respectively.

¹<https://grouplens.org/datasets/movielens/1m/>

²<https://jmcauley.ucsd.edu/data/amazon/>

³<http://snap.stanford.edu/data/loc-gowalla.html>

⁴<https://www.yelp.com/dataset>

Table 3: Overview of performance. The best performing model on each dataset and metric is highlighted in bold, and the second-best model is underlined. The asterisk (*) represents the significance level p -value < 0.05 . Δ represents the difference in performance between AGCF and the second-best model.

Dataset	Metric	BPRMF	NGCF	LightGCN	HGCF	HRCF	HICF	HGCH	GDCF	GraphMoRE	HamGNN	AGCF	$\Delta(\%)$
MovieLens	Recall@10	0.1711	0.1753	0.1835	0.1865	0.1857	0.1861	0.1878	0.1957	<u>0.2043</u>	0.2025	0.2086*	2.10%
	Recall@20	0.2627	0.2663	0.2776	0.2848	0.2829	0.2825	0.2854	0.2965	<u>0.3115</u>	0.3089	0.3193*	2.50%
	NDCG@10	0.2343	0.2386	0.2458	0.2486	0.2471	0.2477	0.2503	0.2628	<u>0.2732</u>	0.2721	0.2808*	2.78%
	NDCG@20	0.2461	0.2511	0.2572	0.2597	0.2578	0.2589	0.2602	0.2720	<u>0.2831</u>	0.2814	0.2922*	3.21%
Amazon-CD	Recall@10	0.0595	0.0742	0.0756	0.0783	0.0812	0.0824	0.0864	0.0871	0.0915	<u>0.0929</u>	0.0984*	5.92%
	Recall@20	0.0868	0.1066	0.1084	0.1098	0.1113	0.1136	0.1172	0.1187	0.1241	<u>0.1272</u>	0.1356*	6.60%
	NDCG@10	0.0338	0.0421	0.0436	0.0442	0.0466	0.0480	0.0495	0.0503	0.0532	<u>0.0546</u>	0.0579*	6.04%
	NDCG@20	0.0409	0.0512	0.0521	0.0535	0.0561	0.0569	0.0587	0.0596	0.0628	<u>0.0641</u>	0.0672*	4.99%
Amazon-Book	Recall@10	0.0606	0.0721	0.0932	0.1037	0.1045	0.1140	0.1172	0.1209	0.1261	<u>0.1294</u>	0.1346*	4.02%
	Recall@20	0.0876	0.1129	0.1362	0.1438	0.1427	0.1551	0.1604	0.1647	0.1738	<u>0.1772</u>	0.1851*	4.46%
	NDCG@10	0.0337	0.0384	0.0569	0.0634	0.0652	0.0713	0.0735	0.0759	0.0802	<u>0.0821</u>	0.0866*	5.48%
	NDCG@20	0.0405	0.0491	0.0689	0.0750	0.0773	0.0826	0.0841	0.0865	0.0911	<u>0.0934</u>	0.0976*	4.49%
Gowalla	Recall@10	0.0854	0.0868	0.1096	0.1106	0.1114	0.1148	0.1217	0.1189	<u>0.1297</u>	0.1275	0.1351*	4.16%
	Recall@20	0.1257	0.1279	0.1588	0.1643	0.1641	0.1646	0.1722	0.1694	<u>0.1813</u>	0.1782	0.1868*	2.87%
	NDCG@10	0.0571	0.0579	0.0749	0.0762	0.0769	0.0793	0.0843	0.0818	<u>0.0912</u>	0.0886	0.0956*	4.82%
	NDCG@20	0.0683	0.0692	0.0886	0.0891	0.0895	0.0921	0.0978	0.0947	<u>0.1046</u>	0.1030	0.1098*	4.97%
Yelp	Recall@10	0.0745	0.0783	0.0849	0.0865	0.0930	0.0945	0.0976	0.1026	0.1054	<u>0.1063</u>	0.1105*	3.95%
	Recall@20	0.1113	0.1116	0.1227	0.1278	0.1362	0.1379	0.1425	0.1471	0.1497	<u>0.1519</u>	0.1588*	4.54%
	NDCG@10	0.0446	0.0456	0.0523	0.0526	0.0564	0.0578	0.0593	0.0616	0.0638	<u>0.0657</u>	0.0685*	4.26%
	NDCG@20	0.0543	0.0554	0.0623	0.0634	0.0661	0.0672	0.0701	0.0732	0.0744	<u>0.0762</u>	0.0809*	6.17%

Table 4: Results of ablation study. The bold numbers denote the most significant performance drop for each metric caused by an ablation.

Model	MovieLens				Amazon-Book				Gowalla			
	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20
AGCF	0.2086	0.2808	0.3193	0.2922	0.1346	0.0866	0.1851	0.0976	0.1351	0.0956	0.1865	0.1098
AGCF w/o G	0.1866	0.2462	0.2793	0.2587	0.1158	0.0724	0.1542	0.0829	0.1186	0.0835	0.1694	0.0959
AGCF w/o M	0.2046	0.2751	0.3122	0.2847	0.1318	0.0832	0.1804	0.0945	0.1308	0.0925	0.1826	0.1050
AGCF w/o S	0.2043	0.2764	0.3049	0.2865	0.1328	0.0855	0.1832	0.0969	0.1320	0.0924	0.1837	0.1066

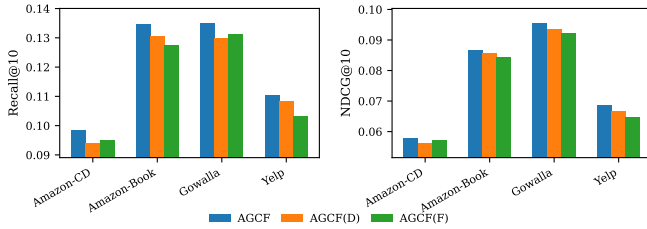


Figure 2: Comparison of the different kinetic energy functions. We compare AGCF against two variants, AGCF(D) and AGCF(F).

5.4 Analysis of the Kinetic Energy Design (RQ3)

To gain a deeper understanding of the rationale behind the structural design of the kinetic energy term K , we conduct an additional comparative analysis. The comparison considers the full AGCF

model, whose kinetic term is induced by a full Riemannian metric, against the following two key variants:

- **AGCF(D) (Diagonal Metric):** In this variant, the Riemannian metric $G(x)$ is constrained to be a diagonal matrix to examine the necessity of the off-diagonal elements in the metric.
- **AGCF(F) (Geometry-free Kinetic):** In this variant, we replace the kinetic energy term K with a generic black-box function, $K_{\text{net}}(x, y)$, implemented by an MLP. This variant completely discards the structure from Riemannian geometry and is designed to test the necessity of the inductive bias itself.

The experimental results in Figure 2 indicate that the performance of AGCF is significantly superior to both variants. The drop observed for AGCF(D) suggests that retaining off-diagonal entries in the adaptive metric is crucial for expressing local geometry and for improving representation quality. Furthermore, the inferior results of AGCF(F) validate that the structured design of the kinetic energy term, derived from Riemannian geometry, is not an unnecessary constraint; on the contrary, this inductive bias, which

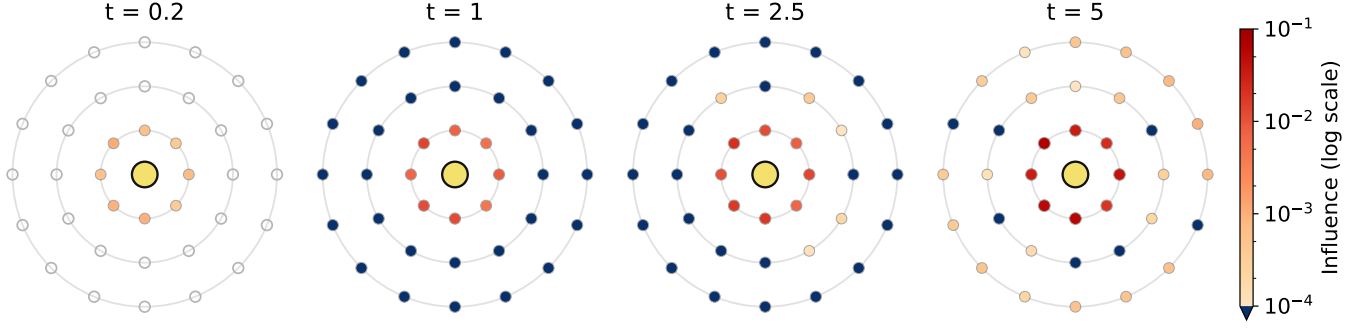


Figure 3: Visualization of influence propagation. The central node is marked in yellow. Nodes in the inner, middle, and outer rings correspond to 1-hop, 2-hop, and 3-hop neighbors, respectively. For visual clarity, only a subset of neighbors is displayed, and the edges between them are omitted. The color of each node, from blue (low) to red (high), represents its influence on the central node at different time snapshots t . Hollow white nodes indicate a numerically negligible influence (close to zero). To highlight significantly activated nodes, all influence values below an activation threshold (10^{-4}) but non-zero are uniformly mapped to dark blue.

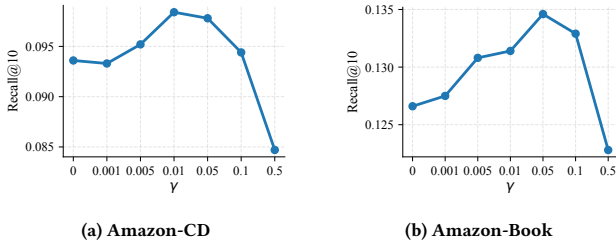


Figure 4: Recall@10 vs. damping coefficient γ . The x-axis shows the tested values of γ , which are arranged at uniform intervals for visual clarity.

determines the geodesic flow, provides a marked performance boost for the model.

5.5 Case Study: Long-Range Influence Propagation (RQ4)

To visually validate the intrinsic mechanism of AGCF for capturing long-range collaborative signals, we design a simulation experiment. This experiment quantifies how the influence of neighbors at different distances from a central node changes over the dynamical evolution time t . We use the MovieLens dataset and select a user as the central node for observation. The influence of a neighbor node v on the central node u at time t is defined as the norm of the Jacobian matrix, $\|\partial x_u(t)/\partial x_v(0)\|$. We compute the influence of each neighbor at several time snapshots. Figure 3 clearly illustrates the propagation of influence over time:

- At an early stage ($t = 0.2$), only the 1-hop neighbors (inner ring) show an initial influence on the central node (yellow).
- As the system evolves to $t = 1.0$, the influence from 1-hop neighbors intensifies, and the influence wave begins to reach the farther neighbors (middle and outer rings), causing them to show a weak signal.

- At $t = 2.5$, the influence has significantly propagated to the 2-hop neighbors, and many of them are lit up as the signal strengthens.
- By $t = 5.0$, the propagation has deepened further: the influence of the 3-hop neighbors (outer ring) is now also enhanced, with many of them being lit up in light red.

This experiment provides visual and quantitative evidence for the theoretical analysis. The observations show that the proposed dynamical formulation naturally propagates information from immediate to distant neighborhoods over time, aligning with the mechanism predicted by the analysis and reinforcing the interpretation of long range effects.

5.6 Impact of the Damping Coefficient (RQ5)

Figure 4 presents the sensitivity analysis of γ , showing a consistent pattern where performance peaks under light damping ($\gamma \in [0.01, 0.05]$). Both purely conservative settings ($\gamma = 0$) and strong damping (e.g., $\gamma = 0.5$) lead to sub-optimal results, indicating that appropriate damping is vital for converging to effective embedding states.

6 Conclusion

This work addresses three fundamental limitations in graph collaborative filtering: geometric homogeneity, tangent space aggregation, and semantic isotropy. To this end, AGCF is introduced as a novel approach based on Hamiltonian dynamics that casts representation learning as a unified physical process evolving on an adaptive geometry. Theoretical analysis and experiments on multiple benchmarks jointly show consistent gains, indicating that the approach is both principled and effective.

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A Proof of Theoretical Results

This section provides a complete and detailed proof for the theoretical results presented in Section 4.3

A.1 Theorem 1

First, we write the entire dynamical system as $\dot{z} = F(z)$, where the state vector is $z = (x, y)$:

$$F(x, y) = \begin{pmatrix} \mathbf{G}(x)^{-1}y \\ -\frac{1}{2}\nabla_x(y^\top \mathbf{G}(x)^{-1}y) - \alpha(\mathbf{M} \otimes \mathbf{S})x - \Gamma \mathbf{G}(x)^{-1}y \end{pmatrix}. \quad (23)$$

Let $f_G(x, y) = \nabla_x(y^\top \mathbf{G}(x)^{-1}y)$. Then the j -th component of this term is given by:

$$[f_G(x, y)]_j = \frac{\partial}{\partial x_j}(y^\top \mathbf{A}(x)y) = y^\top \left(\frac{\partial \mathbf{A}(x)}{\partial x_j} \right) y, \quad (24)$$

where $\mathbf{A}(x) = \mathbf{G}(x)^{-1}$. Using the identity for the derivative of a matrix inverse, we have:

$$\frac{\partial \mathbf{A}(x)}{\partial x_j} = -\mathbf{A}(x) \left(\frac{\partial \mathbf{G}(x)}{\partial x_j} \right) \mathbf{A}(x). \quad (25)$$

Substituting this back, we get:

$$[f_G(x, y)]_j = -(\mathbf{A}(x)y)^\top \left(\frac{\partial \mathbf{G}(x)}{\partial x_j} \right) (\mathbf{A}(x)y). \quad (26)$$

To proof Theorem 1, we need two propositions as follows:

PROPOSITION 3 (LOCAL EXISTENCE AND UNIQUENESS). *Under Assumptions (H1), for any initial condition, the AGCF dynamical system $F(z)$ has a unique local solution.*

According to the Picard-Lindelöf theorem, to prove Proposition 3, we only need to show that the vector field $F(z)$ is locally Lipschitz continuous. We will prove this progressively through a series of lemmas. Throughout the proof, we assume that $z = (x, y)$ belongs to a compact set $\mathcal{K}$.

Step 1: Boundedness and Lipschitz Continuity of $\mathbf{A}(x)$

LEMMA 4. *Under Assumptions (H1), the matrix $\mathbf{A}(x)$ is bounded and locally Lipschitz continuous on any compact set $\mathcal{K}$.*

PROOF. (Boundedness) By Assumption (H1.a), we have $m\mathbf{I} \leq \mathbf{G}(x) \leq M\mathbf{I}$. This implies that the minimum eigenvalue of $\mathbf{G}(x)$ is $\lambda_{\min} \geq m > 0$. Consequently, the maximum eigenvalue of its inverse $\mathbf{A}(x)$ is $\lambda_{\max}(\mathbf{A}(x)) = 1/\lambda_{\min}(\mathbf{G}(x)) \leq 1/m$. Therefore, the norm of $\mathbf{A}(x)$ is bounded on $\mathcal{K}$ by $\|\mathbf{A}(x)\| \leq 1/m$.

(Lipschitz Continuity) First, we show that $\mathbf{G}(x)$ is Lipschitz continuous on $\mathcal{K}$. For any two points $x_1, x_2 \in \mathcal{K}$, since $\mathcal{K}$ is convex⁵, the line segment $x_\theta = x_1 + \theta(x_2 - x_1)$ is contained in $\mathcal{K}$ for all $\theta \in [0, 1]$. By the fundamental theorem of calculus, we have:

$$\mathbf{G}(x_2) - \mathbf{G}(x_1) = \int_0^1 J_G(x_\theta)(x_2 - x_1)d\theta,$$

where $J_G(x_\theta)$ is the Jacobian of $\mathbf{G}$ at the point x_θ . Taking the norm yields:

$$\|\mathbf{G}(x_2) - \mathbf{G}(x_1)\| \leq \int_0^1 \|J_G(x_\theta)\| \|x_2 - x_1\| d\theta.$$

By Assumption (H1.b), the Jacobian norm $\|J_G(x)\|$ is bounded on $\mathcal{K}$, i.e., there exists a constant C_G such that:

$$\sup_{x \in \mathcal{K}} \|J_G(x)\| \leq C_G(\mathcal{K}). \quad (27)$$

Thus,

$$\|\mathbf{G}(x_2) - \mathbf{G}(x_1)\| \leq C_G(\mathcal{K}) \|x_2 - x_1\| \int_0^1 d\theta = C_G(\mathcal{K}) \|x_2 - x_1\|,$$

which proves that $\mathbf{G}(x)$ is Lipschitz continuous with a Lipschitz constant $C_G(\mathcal{K})$.

Next, we show the Lipschitz continuity of $\mathbf{A}(x)$. Using the identity $\mathbf{A}(x_1) - \mathbf{A}(x_2) = \mathbf{A}(x_1)(\mathbf{G}(x_2) - \mathbf{G}(x_1))\mathbf{A}(x_2)$ and taking the norm:

$$\|\mathbf{A}(x_1) - \mathbf{A}(x_2)\| \leq \|\mathbf{A}(x_1)\| \|\mathbf{G}(x_2) - \mathbf{G}(x_1)\| \|\mathbf{A}(x_2)\|.$$

Combining the boundedness of $\mathbf{A}(x)$ and the Lipschitz continuity of $\mathbf{G}(x)$, we obtain:

$$\|\mathbf{A}(x_1) - \mathbf{A}(x_2)\| \leq \frac{1}{m} \cdot (C_G(\mathcal{K}) \|x_1 - x_2\|) \cdot \frac{1}{m} = \frac{C_G(\mathcal{K})}{m^2} \|x_1 - x_2\|.$$

Therefore, $\mathbf{A}(x)$ is Lipschitz continuous on $\mathcal{K}$ with a Lipschitz constant $L_A(\mathcal{K}) = \frac{C_G(\mathcal{K})}{m^2}$. $\square$

Step 2: Lipschitz Continuity of $v(x, y) = \mathbf{A}(x)y$

LEMMA 5. *Under Assumptions (H1), the function $v(x, y) = \mathbf{A}(x)y$ is locally Lipschitz continuous with respect to the state $z = (x, y)$ on any compact set $\mathcal{K}$.*

⁵For any compact set $\mathcal{K}$, we can always find a compact convex subset $\mathcal{K}' \subseteq \mathcal{K}$, on which our Assumption (H1) still holds. For the sake of brevity, we slightly abuse notation and do not distinguish between $\mathcal{K}$ and $\mathcal{K}'$.

PROOF. Consider any two points $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ in $\mathcal{K}$.

$$\begin{aligned} \|v(z_1) - v(z_2)\| &= \|A(x_1)y_1 - A(x_2)y_2\| \\ &\leq \|A(x_1)y_1 - A(x_1)y_2\| + \|A(x_1)y_2 - A(x_2)y_2\| \\ &\leq \|A(x_1)\| \|y_1 - y_2\| + \|A(x_1) - A(x_2)\| \|y_2\|. \end{aligned}$$

Since $\mathcal{K}$ is compact, y is bounded, i.e., there exists $R_y(\mathcal{K})$ such that $\|y\| \leq R_y(\mathcal{K})$. Combining with the results from Lemma 4, we have:

$$\|v(z_1) - v(z_2)\| \leq \frac{1}{m} \|y_1 - y_2\| + \frac{C_G(\mathcal{K})R_y(\mathcal{K})}{m^2} \|x_1 - x_2\|, \quad (28)$$

which is the Lipschitz condition with respect to $z = (x, y)$. $\square$

Step 3: Lipschitz Continuity of the Geometric Force $f_G(x, y)$

LEMMA 6. Under Assumptions (H1), the geometric force term $f_G(x, y)$ is locally Lipschitz continuous with respect to the state $z = (x, y)$ on any compact set $\mathcal{K}$.

PROOF. According to $v = A(x)y$, the i -th component of the geometric force can then be rewritten as $[f_G(x, y)]_i = -v^\top \frac{\partial G(x)}{\partial x_i} v$. We aim to show that there exists a constant $L_f(\mathcal{K})$ such that for any $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2) \in \mathcal{K}$, the inequality $\|f_G(z_1) - f_G(z_2)\| \leq L_f(\mathcal{K}) \|z_1 - z_2\|$ holds. Let $v_1 = A(x_1)y_1$ and $v_2 = A(x_2)y_2$. We analyze the absolute difference of the i -th component:

$$|[f_G(z_1)]_i - [f_G(z_2)]_i| = |v_2^\top \frac{\partial G(x_2)}{\partial x_i} v_2 - v_1^\top \frac{\partial G(x_1)}{\partial x_i} v_1|. \quad (29)$$

We decompose the difference into three parts:

$$\begin{aligned} &\leq \underbrace{|v_2^\top \frac{\partial G(x_2)}{\partial x_i} v_2 - v_1^\top \frac{\partial G(x_2)}{\partial x_i} v_2|}_{(T1)} + \underbrace{|v_1^\top \frac{\partial G(x_2)}{\partial x_i} v_2 - v_1^\top \frac{\partial G(x_1)}{\partial x_i} v_2|}_{(T2)} \\ &\quad + \underbrace{|v_1^\top \frac{\partial G(x_1)}{\partial x_i} v_2 - v_1^\top \frac{\partial G(x_1)}{\partial x_i} v_1|}_{(T3)}. \end{aligned}$$

We bound each term separately:

- T1: $|(v_2 - v_1)^\top \frac{\partial G(x_2)}{\partial x_i} v_2| \leq \|v_2 - v_1\| \left\| \frac{\partial G(x_2)}{\partial x_i} \right\| \|v_2\|$.
- T2: $|v_1^\top \left(\frac{\partial G(x_2)}{\partial x_i} - \frac{\partial G(x_1)}{\partial x_i} \right) v_2| \leq \|v_1\| \|v_2\| \left\| \frac{\partial G(x_2)}{\partial x_i} - \frac{\partial G(x_1)}{\partial x_i} \right\|$.
- T3: $|v_1^\top \frac{\partial G(x_1)}{\partial x_i} (v_2 - v_1)| \leq \|v_1\| \left\| \frac{\partial G(x_1)}{\partial x_i} \right\| \|v_2 - v_1\|$.

On a compact set $\mathcal{K}$, the state vectors x and y are bounded. From Lemma 4, $A(x)$ is bounded, which implies that the velocity $v = A(x)y$ is also bounded by a constant $R_v(\mathcal{K})$. From Assumption (H1b), the derivative $\partial G/\partial x$ is Lipschitz continuous. This means there exist constants $C_G(\mathcal{K})$ and $L_G(\mathcal{K})$ such that:

$$\begin{aligned} \|\partial G(x)/\partial x_i\| &\leq C_G(\mathcal{K}), \\ \|\partial G(x_1)/\partial x_i - \partial G(x_2)/\partial x_i\| &\leq L_G(\mathcal{K}) \|x_1 - x_2\|. \end{aligned}$$

Furthermore, from Lemma 5 (the Lipschitz continuity of $v(x, y)$), there exists a constant $L_v(\mathcal{K})$ such that $\|v_1 - v_2\| \leq L_v(\mathcal{K}) \|z_1 - z_2\|$.

Substituting these bounds into the three terms yields:

$$\begin{aligned} T1 &\leq (L_v(\mathcal{K}) \|z_1 - z_2\|) R_v(\mathcal{K}), \\ T2 &\leq R_v(\mathcal{K})^2 (L_G(\mathcal{K}) \|x_1 - x_2\|), \\ T3 &\leq R_v(\mathcal{K}) C_G(\mathcal{K}) (L_v(\mathcal{K}) \|z_1 - z_2\|). \end{aligned}$$

Combining the Bounds. Summing the bounds for all terms, we find that each component of $f_G(z)$ is bounded by a constant multiple of $\|z_1 - z_2\|$. Therefore, the vector $f_G(z)$ is locally Lipschitz continuous. $\square$

Step 4: Lipschitz Continuity of the Full Dynamics $F(z)$

We now introduce two general lemmas regarding Lipschitz functions.

LEMMA 7 (SUM OF LIPSCHITZ FUNCTIONS). If functions $f : U \rightarrow \mathbb{R}^n$ and $h : U \rightarrow \mathbb{R}^n$ are Lipschitz continuous on a set $U \subseteq \mathbb{R}^m$ with Lipschitz constants L_f and L_h respectively, then their sum $f + h$ is also Lipschitz continuous on U with a constant of at most $L_f + L_h$.

PROOF. By definition and the triangle inequality, for any $u, v \in U$:

$$\begin{aligned} \|(f(u) + h(u)) - (f(v) + h(v))\| &= \|(f(u) - f(v)) + (h(u) - h(v))\| \\ &\leq \|f(u) - f(v)\| + \|h(u) - h(v)\| \\ &\leq (L_f + L_h) \|u - v\|. \end{aligned} \quad \square$$

LEMMA 8 (LIPSCHITZ CONTINUITY OF VECTOR COMPONENTS). A vector function $F : U \rightarrow \mathbb{R}^{\sum h_i}$, $F = (F^{(1)}, \dots, F^{(n)})$, is Lipschitz continuous on a set $U \subseteq \mathbb{R}^m$ if and only if each of its component functions $F^{(i)} : U \rightarrow \mathbb{R}^{h_i}$ is Lipschitz continuous.

PROOF. ($\Rightarrow$) If F is Lipschitz with constant L_F , then for any component i :

$$|F^{(i)}(u) - F^{(i)}(v)| \leq \|F(u) - F(v)\| \leq L_F \|u - v\|,$$

so each component $F^{(i)}$ is Lipschitz.

($\Leftarrow$) If each component $F^{(i)}$ is Lipschitz with constant L_i , then:

$$\|F(u) - F(v)\|^2 = \sum_{i=1}^n (F^{(i)}(u) - F^{(i)}(v))^2 \leq \left(\sum_{i=1}^n L_i^2 \right) \|u - v\|^2.$$

Thus, F is Lipschitz with a constant $L_F = \sqrt{\sum L_i^2}$. $\square$

We now return to the proof for $F(z)$. Its two block components are analyzed separately.

- The first block, $G(x)^{-1}y$, is Lipschitz by Lemma 5.
- The second block is a sum of three terms: $-\frac{1}{2}f_G(x, y)$, which is Lipschitz by Lemma 6; $-\alpha(M \otimes S)x$, which is linear and thus globally Lipschitz; and $-\Gamma(G(x)^{-1}y)$, which is Lipschitz by Lemma 5 and the fact that Γ is a constant matrix.

According to Lemma 7, the sum of Lipschitz functions is Lipschitz, we conclude that both block components of $F(z)$ are Lipschitz, and thus $F(z)$ as a whole is Lipschitz continuous by Lemma 8. The Picard-Lindelöf theorem guarantees the existence of a maximal time of existence, $T_{\max} \in (0, \infty]$, and a unique solution $z(t)$ on the time interval $[0, T_{\max})$, Proposition 3 holds.

PROPOSITION 9 (BOUNDEDNESS OF SOLUTIONS). Under Assumptions (H1), any local solution $z(t)$ guaranteed by Proposition 3 is bounded on its entire interval of existence.

LEMMA 10 (EQUIVALENCE OF ENERGY AND NORM). Under Assumption (H1.a), there exist $c_1, c_2 > 0$ such that for any state $z(t)$, the following inequality holds:

$$c_1 \|z(t)\|^2 \leq H(x, y) \leq c_2 \|z(t)\|^2.$$

PROOF. The Hamiltonian is $H(x, y) = \frac{1}{2}y^\top G(x)^{-1}y + \frac{\alpha}{2}x^\top (\mathbf{M} \otimes \mathbf{S})x$.

By Assumption (H1.a), the kinetic term is therefore bounded by:

$$\frac{1}{2M}\|y\|^2 \leq \frac{1}{2}y^\top G(x)^{-1}y \leq \frac{1}{2m}\|y\|^2. \quad (30)$$

For the potential term, we have $\lambda_{\min}(\mathbf{M} \otimes \mathbf{S})\|x\|^2 = \lambda_{\min}(\mathbf{M})\lambda_{\min}(\mathbf{S})\|x\|^2 \leq x^\top (\mathbf{M} \otimes \mathbf{S})x \leq \lambda_{\max}(\mathbf{M} \otimes \mathbf{S})\|x\|^2 = \lambda_{\max}(\mathbf{M})\lambda_{\max}(\mathbf{S})\|x\|^2$. Combining these, we get:

$$\begin{aligned} H(x, y) &\leq \frac{1}{2m}\|y\|^2 + \frac{\alpha}{2}\lambda_{\max}(\mathbf{M})\lambda_{\max}(\mathbf{S})\|x\|^2 \\ &\leq \max\left\{\frac{1}{2m}, \frac{\alpha}{2}\lambda_{\max}(\mathbf{M})\lambda_{\max}(\mathbf{S})\right\}(\|x\|^2 + \|y\|^2) = c_2\|z\|^2, \end{aligned}$$

where $c_2 = \max\{\frac{1}{2m}, \frac{\alpha}{2}\lambda_{\max}(\mathbf{M})\lambda_{\max}(\mathbf{S})\}$. Also,

$$\begin{aligned} H(x, y) &\geq \frac{1}{2M}\|y\|^2 + \frac{\alpha}{2}\lambda_{\min}(\mathbf{M})\lambda_{\min}(\mathbf{S})\|x\|^2 \\ &\geq \min\left\{\frac{1}{2M}, \frac{\alpha}{2}\lambda_{\min}(\mathbf{M})\lambda_{\min}(\mathbf{S})\right\}(\|x\|^2 + \|y\|^2) = c_1\|z\|^2, \end{aligned}$$

where $c_1 = \min\{\frac{1}{2M}, \frac{\alpha}{2}\lambda_{\min}(\mathbf{M})\lambda_{\min}(\mathbf{S})\}$. $\square$

LEMMA 11 (ENERGY DECAY). *Along any solution trajectory of the AGCF dynamical system, the total energy H is non-increasing with respect to time. That is:*

$$\frac{dH}{dt} \leq 0.$$

PROOF. We compute the time derivative of H along the trajectories of the system:

$$\frac{dH}{dt} = \left(\frac{\partial H}{\partial x}\right)^\top \dot{x} + \left(\frac{\partial H}{\partial y}\right)^\top \dot{y}.$$

Substituting Hamilton's equations, $\dot{x} = \partial H / \partial y$ and $\dot{y} = -\partial H / \partial x - \Gamma(\partial H / \partial y)$, yields:

$$\begin{aligned} \frac{dH}{dt} &= \left(\frac{\partial H}{\partial x}\right)^\top \left(\frac{\partial H}{\partial y}\right) + \left(\frac{\partial H}{\partial y}\right)^\top \left(-\frac{\partial H}{\partial x} - \Gamma \frac{\partial H}{\partial y}\right) \\ &= \left(\frac{\partial H}{\partial x}\right)^\top \left(\frac{\partial H}{\partial y}\right) - \left(\frac{\partial H}{\partial y}\right)^\top \left(\frac{\partial H}{\partial x}\right) - \left(\frac{\partial H}{\partial y}\right)^\top \Gamma \left(\frac{\partial H}{\partial y}\right). \end{aligned}$$

Since $(\partial H / \partial x)^\top (\partial H / \partial y)$ is a scalar, it is equal to its own transpose, $(\partial H / \partial y)^\top (\partial H / \partial x)$. Thus, the first two terms cancel out, leaving:

$$\frac{dH}{dt} = -\left(\frac{\partial H}{\partial y}\right)^\top \Gamma \left(\frac{\partial H}{\partial y}\right).$$

As the damping matrix Γ is positive semi-definite, the quadratic form $v^\top \Gamma v \geq 0$ for any vector v . Therefore, we conclude that $dH/dt \leq 0$. $\square$

The fact that $dH/dt \leq 0$ implies $H(t) \leq H(0)$ for all t in the interval of existence. Combining this with Lemma 10, we have:

$$c_1\|z(t)\|^2 \leq H(t) \leq H(0).$$

This implies that $\|z(t)\|^2 \leq \frac{1}{c_1}H(0)$, which shows that the solution $z(t)$ is uniformly bounded on its maximal interval of existence $[0, T_{\max})$. Thus, Proposition 9 holds.

The proof of this theorem is a direct consequence of the preceding propositions. By Proposition 3, we know that a unique local solution exists on an interval $[0, T_{\max})$. Assume for contradiction that $T_{\max} < \infty$. By the extension theorem for ODEs, if the maximal time of existence is finite, the solution must "blow up", meaning its norm must diverge to infinity as $t \rightarrow T_{\max}$, i.e., $\|z(t)\| \rightarrow \infty$. However, we have shown in Proposition 9 that the solution $z(t)$ is bounded for all $t \in [0, T_{\max})$. This is a contradiction. The solution cannot blow up in finite time. Therefore, the only possibility is that $T_{\max} = \infty$. Thus, the unique solution exists for all time $t \in [0, \infty)$. Theorem 1 holds.

A.2 Theorem 2

As stated in the main text, under the simplifying condition that the metric matrix G is constant, the first-order dynamical system (14) can be exactly rewritten as a standard second-order linear ordinary differential equation with constant coefficients:

$$\ddot{x}(t) + \mathbf{D}\dot{x}(t) + \Omega x(t) = 0,$$

where the constant matrices are $\mathbf{D} = \mathbf{G}^{-1}\Gamma$ and $\Omega = \alpha\mathbf{G}^{-1}(\mathbf{M} \otimes \mathbf{S})$.

By repeatedly differentiating the second-order ODE, we can compute the higher-order derivatives at $t = 0$:

- $x^{(2)}(0) = -\mathbf{D}\dot{x}(0) - \Omega x_0$.
- $x^{(3)}(0) = (\mathbf{D}^2 - \Omega)\dot{x}(0) + \mathbf{D}\Omega x_0$.
- $x^{(4)}(0) = (-\mathbf{D}^3 + \mathbf{D}\Omega + \Omega\mathbf{D})\dot{x}(0) + (-\mathbf{D}^2\Omega + \Omega^2)x_0$.

By observing the expansion of the derivatives, one can see that for any $k \geq 1$, the unique term in the expansion of $x^{(2k)}(0)$ that reaches the k -th power of Ω is the one generated by applying the $-\Omega$ operator k consecutive times, which is $(-\Omega)^k x_0$. Therefore, the Taylor expansion of $x(t)$ necessarily contains a subsequence of these "skeleton" terms corresponding to the even powers of t :

$$x(t) = \dots + \frac{t^2}{2!}(-\Omega)x_0 + \dots + \frac{t^{2k}}{(2k)!}((-1)^k \Omega^k)x_0 + \dots$$

Next, we establish the formal connection between the algebraic term Ω and the graph topology. Since G is a constant block-diagonal matrix, $G = \text{blockdiag}(G_1, \dots, G_N)$, its $d \times d$ sub-block Ω_{ij} is given by:

$$\Omega_{ij} = \alpha G_i^{-1} M_{ij} S.$$

This means the block-sparse structure of Ω is identical to that of the structural matrix $\mathbf{M}$.

In summary, the term $(-\Omega)^k x_0$ in the Taylor series captures how the initial information of k -hop neighbors is injected into the current node via graph paths. As t increases, the t^{2k} factor makes the influence of larger hops more prominent.